

## Pure Substance

**Definition.** A pure substance is a system composed of a single chemical species, exhibiting uniform and invariant chemical composition throughout its mass, whether it exists in one phase or coexists in several. Formally, if  $N$  denotes the total number of moles and  $N_i$  the moles of chemical component  $i$ , a pure substance satisfies  $N_i = N$  for exactly one  $i$ , and  $N_j = 0$  for all  $j \neq i$ . Crucially, a pure substance need not be chemically homogeneous in phase: liquid water in equilibrium with its vapor is a pure substance, since both phases consist entirely of  $\text{H}_2\text{O}$  molecules, even though they differ in molar volume and density.

**Worked example.** Consider a rigid, sealed container holding 2 kg of water at  $100^\circ\text{C}$  and 1 atm, existing partly as liquid and partly as vapor. Although two phases are present, the system is a pure substance because its chemical composition is uniform: every parcel of matter, liquid or vapor, is  $\text{H}_2\text{O}$ . If instead the container held a mixture of liquid water and liquid oil, the system would *not* be pure, despite being single-phase in appearance, since the composition varies with the identity of the molecular species present.

**Key theorem (Gibbs Phase Rule specialization).** For a pure substance ( $C = 1$  component), the Gibbs phase rule,

$$F = C - P + 2,$$

reduces to  $F = 3 - P$ , where  $F$  is the number of independent intensive variables (degrees of freedom) and  $P$  is the number of coexisting phases. Consequently, a single-phase pure substance has  $F = 2$  (e.g.,  $T$  and  $v$  fix the state), while a two-phase mixture has  $F = 1$  (temperature alone fixes the saturation pressure and vice versa).

**Lab cell (SymPy).**

```
import sympy as sp

C, P, F = sp.symbols('C P F', positive=True, integer=True)
gibbs_rule = sp.Eq(F, C - P + 2)
```

```
pure_single_phase = gibbs_rule.subs({C: 1, P: 1})  
pure_two_phase = gibbs_rule.subs({C: 1, P: 2})  
  
print(sp.simplify(pure_single_phase))  
print(sp.simplify(pure_two_phase))
```